

Input: A single, long prompt
Containing: 1. A puzzle in an unknown token language, 2. Grammar rules for the language, and 3. An example solution using a backtracking algorithm

LLM

Internal Tasks for the LLM

Task 1: Grammar Learning
Interpret token language rules

Task 2: Algorithm Learning
Extract backtracking structure from the example

Task 3: Code Implementation
Combine grammar and algorithm to implement a Python solver

Output: Final Python solver code

Automated Verification by Pytest

Runtime Error (e.g.,
KeyError)

Result: Failure